

AI-Assisted AutoLISP in AutoCAD: A Practical Workflow for Non-Programmers

Abstract

Most AutoCAD teams use powerful Autodesk tools every day, but very few engineers ever touch AutoLISP. In our railway engineering department, that meant many small, repetitive tasks stayed manual unless a specialist development team stepped in. In this session, we will demonstrate how we leveraged this situation through AI-assisted AutoLISP workshops for engineers with little or no coding experience. Attendees will learn how to identify suitable automation candidates, prompt AI to generate useful routines, test and refine output, and introduce light-weight governance such as version control and validation. The result of implementing AI-Assisted AutoLISP routines is faster, streamlined engineering work, more self-sufficient teams, and a practical path from AI curiosity to repeatable workflow improvement.

Across our department, and increasingly across our wider company, engineers work daily with Autodesk tools such as AutoCAD, Civil 3D, and Inventor. The overall level of product expertise is high, but one capability has remained largely untapped: AutoLISP in AutoCAD. Only a small number of people had ever written a routine, and many colleagues had never even heard of it.

My team, Rail Digital Engineering, has been building software for our engineers for years. That approach works very well, but it does not always address the small, highly specific tasks that can be automated quickly and that immediately improve day-to-day work. Those smaller opportunities often remain untouched because they are not large enough to justify a formal development effort.

That changed with the arrival of accessible AI tools. We realized that designers with little or no programming background could start creating useful AutoLISP routines with AI support. Instead of waiting for a central team to build every solution, they can improve their own workflows directly. What they needed was not traditional software training, but awareness, a practical method, and a guided workshop in which they could build routines themselves without needing to become programmers first.

The results were remarkable. The sessions were highly engaging, and participants achieved more in a short time than many of them thought possible. More importantly, the workshop did not just create enthusiasm; it changed behavior. For many colleagues, it became the starting point for applying AI-assisted automation in their daily work. Once people understand the process, learn a few smart prompting and validation habits, and receive minimal guidance, they can continue on their own.

At the same time, successful adoption requires more than enthusiasm. If people want to share code, they need to understand version control. If they use AI to generate routines, they also need to test and validate those routines properly. Even if they are not writing every line of code by hand, they still need confidence that the output is reliable, safe, and maintainable. That combination of empowerment, governance, and practical workflow improvement is what makes this topic so relevant.

This session could take one of two forms: it could be a case study that shows how we introduced AI-assisted AutoLISP, what worked, and how others can replicate it in their own organizations; or it could be a live hands-on workshop in which attendees experience the process for themselves and leave with a working routine, a repeatable method, and a clear understanding of the guardrails needed to scale it responsibly.